NGA CLOUD GUIDANCE: INTRODUCTION

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Preamble

IC ITE Initiative

The Intelligence Community has undertaken the IC Information Technology Enterprise (IC ITE) initiative as a platform for strengthening mission integration across the IC. IC ITE has designated service providers for cloud and common support services to "do in common what is commonly done." NGA has supported its "all in" position by adopting community-provided services and GEOINT services for the community in a cloud-based enterprise.

The IC has established greater interoperability and efficiency within the community, and implemented an enterprise that will forge enduring synergies between agencies along with greater mission support and effectiveness. IC ITE has reached the stage where its enterprise services are fulfilling that sustainability goal of enabling IC interoperability and efficiency with the benefit of greater IC mission support.

NGA's Future is in the Cloud

NGA's core mission is about understanding the world, and conveying that understanding, in spatio-temporal terms. All agencies are increasingly relying upon geospatial capabilities in the cloud's global topology. NGA is pursuing new analytical perspectives as identified in the GEOINT 2022 CONOPS. NGA will provide our GEOINT services for enterprise consumption and forge agreements with commercial providers offering advantageous GEOINT capabilities.

GEOINT is a vital part of mission support, and we can only do our part if geospatial content and services exist with a global footprint. NGA's services must be in the cloud so that the IC, DoD and our Allies can take advantage of the cloud's ever presence. Our ability to realize the objectives of the GEOINT 2022 CONOPS is dependent on operating in the cloud and maximizing access to geospatial data and big data analytics to support the mission.

By setting the GEOINT 2022 goals, we will challenge ourselves to think differently about our future. The time to achieve these goals does not afford us the luxury of waiting, we must progressively implement those capabilities in the cloud.

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Cloud Guidance: Introduction

1.0 Introduction

1.1 Applicability

This Cloud Guidance document provides NGA Program Managers (PMs), developers, engineers, and organizational managers with a basic understanding of actions, constraints, and expectations of operations in the cloud environment.

This document itself is not contractual, but provides guidance to apply to contractual actions for cloud-based activities. The Annexes and White Papers are linked to references that provide more detailed technical implementation details important to architects, engineers, and developers.

For purposes of this guidance, an "activity" refers to any project, program, system, entity, instance, IT asset, or data that exists at NGA. It encompasses all software; hardware; purchased and NGA-produced data; and mission and business applications associated with the activity, whether funded by NGA or managed by NGA for others (e.g., when we receive funds from others) to provide services on their behalf. Collectively, these terms are referred to as an "activity".

The mission objectives discussed in the GEOINT 2022 Concept of Operations (CONOPS) will be achieved as we operate in the cloud. NGA will endeavor to operate all its enterprise activities in the cloud. NGA's move to the cloud is one of many transformational initiatives that will occur on that path with the cloud being the foundation to enable that global mission support vision.

1.2 Cloud Strategy

Our success can be measured by how well we deliver the objective outcome. Mission capability analyses for GEOINT 2022 indicate that most of our legacy activities do not align with the path forward. The cloud strategy is to focus resources on efforts that will drive NGA forward. NGA's scarce resources are best focused in the development of capabilities that would reset NGA value proposition to its stakeholders. This premise is based on the reality of scarce resources and three key assumptions:

- **"What got us here will not get us there":** Director Cardillo's statement in the NGA's GEOINT 2022 CONOPS acknowledges that our future is not a continuation of our past. The 2022 CONOPS describes three phases in the journey forward namely the modernization, automation, and anticipatory analysis. In general, cloud migration provides impetus for modernization, which enables automation of GEOINT workflows and data processing to increase our resources and focus on anticipatory analysis. The recent planning guidance calls for the automation of 75% of what NGA does today and the anticipatory phase is expected to be based on profile modeling and analytics to anticipate an adversary's next move. These two last phases will require innovation and substantial development resources.

- **Refactoring and/or re-engineering applications take considerable resources**: Gartner's cloud migration industry benchmarks call for 24 months for a lift and shift, 36 months for refactoring and 48+ months for re-engineering the solution. The refactor benchmarks were verified by NGA when it evaluated the migration of key programs. NGA's foundational phase centered on "smart" lift and shift. The "smart" part of the approach generated cloud guidance and support (i.e., enterprise services, IaaS, PaaS, DevOps, and REvAMP).
- **Success is realizing the end goal, not necessarily meeting intermediate milestones**: Cloud migration is the basis for accomplishing automation, which allows resources to focus on the end goal of anticipatory analysis. In this light, cloud migration should be done to the level which is effective in the path forward to support the GEOINT 2022 objectives. This progressive mindset is reinforced by the fact that more than 50% of our current efforts will not migrate.

Given that the realization of the 2022 CONOPS requires substantial effort and that NGA's future is not a projection of its past, investments in legacy activities should be minimized in lieu of investment in new mission-focused capabilities. Cloud employment to meet GEOINT 2022 goals advocate a strategy pivot to align ongoing activities and resources in the following manner:

- Tier 1 – New efforts aligned with the 2022 CONOPS

These efforts will follow cloud guidance to include being cloud native, leveraging PaaS, following digital transformation processes (e.g., DevOps) and separating data from applications.

- Tier 2 – Key activities (i.e., activities that persist to facilitate the 2022 CONOPS)

These efforts will be refactored, if needed, to migrate and follow cloud guidance.

- Tier 3 – All other activities selected for migration

These efforts will not be refactored unless there is a benefit/return on investment as determined by a business case and approved by NGA.

The cloud strategy's pivot is to pursue a Tier 1 approach, since it maximizes investment based on Return on Investment (ROI). The ROI criteria reverses the previous approach by moving from a "migrate-yes", to a "migrate-no" default. It is based on the acknowledgement that the future is not a continuation of the past; and, recognizes the resources and investment needed to develop the required GEOINT 2022 capabilities. Applying ROI necessitates determining mission value via business cases and resets development to follow cloud guidance. Business cases can be explored if improving IT efficiencies outweigh the cost incur. Given the capabilities needed to realize the GEOINT 2022 end goal - *Why spend the time and resources on changes that do not move us forward?*

NGA has an established cloud migration process, the enterprise services suite is in place, and the foundational cloud engineering and support enablers (e.g., IaaS, PaaS, DevOps, REvAMP, OpenDataStore) have been initiated.

Cloud activities will pursue use of "best practices", leverage enterprise services and capabilities, and pursue cost avoidance where possible. By engaging in refactoring efforts that provide proven ROI and proactively retiring legacy activities that do not align with our future, NGA can maximize use of its limited resources towards forging its path forward to realize the 2022 CONOPS.

2.0 Guidance

The Cloud Guidance series applies to all community and public clouds available to NGA for use at any given point in time. The CIO-T will update this guidance document, as required.

Guidance: *There shall be a cloud plan for the agency.*

The NGA Cloud Guidance document is a "living" planning document, which will evolve as we progress on that path to GEOINT 2022. CIO-T will coordinate the integrated cloud plan for the agency and address any cloud implementation issues for NGA's key components.

The NGA Cloud Guidance contains binding guidance and requirements for PMs and an activity's shareholders to apply to cloud-based implementation and operations. The NGA Cloud Strategy is a higher level document that outlines our strategic intent. Wherever there is disparity between these two documents, the NGA Cloud Guidance shall take precedence.

Guidance: *Cloud development and operations shall employ standardize the tools, methods, and templates.*

NGA seeks to maximize use of standardized tools, methods, and templates. Standardization strives to improve responsiveness, security, and integration at the enterprise level. New tools and services that provide greater capability will be incorporated and become part of the "standardized" suite, as they become available. NGA vetted, accredited cloud-based tools can be found at the NGA Cloud Information Portal. Before building enterprise services, survey other agencies to determine if we can adopt or adapt any services or tools they may have already fielded.

Guidance: *Cloud development and operations shall apply the following guidance:*

- *Register "gold" data in ATLAS's activities listing.*
  The Data Inventory Portal hosts data migration information in support of NGA's cloud governance process. "Gold" data is the most current information about an activity. All activities should maintain their gold data in the ATLAS and ensure that the data is sharable and up to date.
- *Establish a date to retire legacy activities in place.*

Legacy or superseded activities shall retire no later than contract end date or ISP recap date or migrate to the cloud. All activities will follow the processes in the PM's Guide to Cloud Migration via the Cloud Adoption Template (CAT).

- *Disposition data in accordance with Annex G: Cloud Data Guidance.*
- *Adopt required enterprise services, as specified in Annex D: Enterprise Services Guidance.*
  Enterprise services will be provided as government furnished property to program offices. In some cases, activities will have to switch from their current service to the enterprise service.
- *Follow security guidance, as provided in Annex I: Cloud Security Guidance.*
- *Follow DevOps guidance, as provided in Annex E: Lifecycle Guidance.*
  Annex E prescribes the approved tools, methods, and control points for conducting DevOps activities for NGA.
- *Utilize pre-approved design templates, as documented in Annex F: Design Guidance.*
  New design templates or additional capabilities will be provided as required.
- *Maximize enterprise services' operational availability and durability in cloud operations.*
  The inherent design of the cloud services, can significantly increase operational availability. Activities must seek to maximize use of inherently durable cloud resources (e.g., S3). For applications that are expected to be highly available, activity owners must deploy such applications across Availability Zones (AZs) and employ load balancing. Activities must seek to use auto-scaling groups to respond to changes in demand in lieu of reserving capacity designed to peak load. For database operations, databases considered critical to application function must be deployed in multi-AZ failover configuration using either an activity defined construct or through the use of cloud-native database managed services.
- *Require unlimited data rights in contract language.*
  Allow non-NGA users and agencies to utilize GEOINT services and content and reuse code. Wherever possible, require that programs not use proprietary or fee-bearing software when other options are available that will improve the ability to share product and source code.
- *Treat cloud guidance as a contract requirement for all contracted activities.*
  Incorporate the intent of this guidance into contract language and direction as needed for all activities.
- *New investments are to be cloud native, as identified by the "Twelve Factor App Principles".*

  New investments are to be based on the prioritized set of tradeoffs. The intent is to first leverage capabilities/services available from either industry and/or other agencies to include those offered in the IC Marketplace and AppsMall. The needed functional capabilities and/or services should follow the following criteria, in priority order:

1. Adopt: Adopt capabilities and/or services available from the community we operate in to include the intelligence, defense and civil communities.
2. Adopt and modify: If the development and/or mission capability and/or services is not completely suitable, we shall work with the provider to adopt and modify it to evolve and expand the original capability and/or service.
3. Acquire and augment: If unable to adopt or adopt and modify the intent would be to acquire a core capability/service and augment as needed to meet the stated requirement.
4. Develop and publish: If unable to adopt and/or adopt and modify, development should follow the cloud guidance and seek to make the capabilities/services available to the community via cloud services.

Note**:** Investments using lower priority precepts should make available the analysis negating the use of higher priority criteria.

- *Exceptions to cloud guidance shall be handled through a governance process.*
  All guidance must allow for exceptions and NGA's cloud guidance. The CIO-T will handle requests for exceptions (i.e., cloud-based implementation) via a review and waiver adjudication process. PMs must submit a waiver form and supporting documentation to the Cloud Engineering Panel POCs. Additional approvals may be required (e.g., MIB). The minimum criteria to request an exception are:
  - Data-driven evidence that the cloud environment as it currently exists cannot support the activity.
  - Evidence that the activity has made every effort to request the necessary Cloud Service Provider (CSP) support.

3.0 Annex Summaries

The NGA cloud guidance is provided through a series of Annexes that address the topic specifics, processes and procedures to be met as NGA activities are established in the cloud. This document and its supporting Annexes can be found at the Cloud Information Portal. The site includes a Key Documents folder along with the Cloud Guidance folder holding the Annexes. Appendix A - Glossary of Abbreviations and Acronyms is a consolidated listing for all the Annexes. Summary information to the current version of the Annex materials follow.

Annex A: IC ITE Services and Providers
This Annex provides a brief introduction to IC ITE services, with links to sources of further information.

Annex B: GEOINT Services Designation
The DNI on 4 December 2014 designated NGA as a provider of a "Service of Common Concern" for common GEOINT services on behalf of the Intelligence Community. This Annex provides a copy of the designation memorandum.

Annex C: Deleted
Cloud Adoption Management Group (CAMG) Charter deleted.

Annex D: Enterprise Services Guidance
Enterprise Services consist of services within two broad categories: mandatory and optional as listed in the NGA Cloud Enterprise Services Catalog which is outlined in the Annex. The catalog is updated regularly with additional service offerings. Activity owners are encouraged to check for updates.

Annex E: Lifecycle Guidance
This Annex describes NGA's lifecycle processes as it relates to the DevOps process. The annex provides guidance for a Software Development Lifecycle centered on the principles of Culture, Automation, Measurement and Sharing (CAMS). It is important to note that Lifecycle is independent of Design Pattern, meaning that the successful employment of a software development approach such as that described by DevOps does not rely on use of a particular design pattern.

Annex F: Design and Implementation Patterns Guidance
This Annex is to introduce design patterns for all activities to follow within the cloud. Detailed information on each individual pattern evolves as best practices emerge and lessons learned are shared from experiences gained. This Annex provides a basic understanding of design patterns, with detailed information provided in supplementary materials relevant to a particular activity's needs.

A design pattern is a generalized approach to solving activity deployment use cases. As implementation progresses the design patterns will be augmented with detailed artifacts enabling repeatable implementation.

Enterprise services and maturity of some platform services are required to meet the intent of the design patterns. The Annex describes available enterprise and platform services for making informed decisions regarding pattern selection.

Annex G: Cloud Data Guidance
This Annex contains both prescriptive and recommended data architecture considerations for an activity's cloud implementation. Activity owners will need to be particularly familiar with the content migration approval process. Classes of data, including that which constitutes mission data and that which is specific only to an application's use is defined.

Annex H: Reserved for future use

Annex I: Cloud Security Guidance
This Annex provides activity owners with the major tenets of security in the cloud, with impacts to the enterprise architecture, cloud security service offerings, and inheritable Risk Management Framework (RMF) controls, and IC compliance standards. The guidance addresses key security requirements and offers recommended best practices and defines the security architecture, tools, and processes required to achieve success.

Annex J: Cloud Architecture Guidance
This Annex provides developers with an overview of NGA's architecture approach with respect to operating to the cloud. In addition, this document provides sufficient context to interpret NGA cloud architecture artifacts. All of the artifacts are identified to provide insight into adopting cloud solutions to support NGA missions. The tables within the Annex present a summary of architecture deliverables with a brief explanation of their benefit.

Annex K: Business Models Guidance
This Annex provides guidance on key NGA and IC business practices and processes managing resources in support of cloud-based operations. Specific IT cloud practices that include:

- Financial planning
- Cloud provisioning
- Cloud cost management
- Software licensing scenarios
- Acquisition considerations

The Annex overviews standard business processes that enable optimized use of cloud-based resources.

Appendix A - Glossary of Abbreviations and Acronyms

Acronym	Description
A&A	Assessment and Authorization
A2SI	Agency Agility Strategic Initiative
AAS	Authoritative Attribute Stores
ABAC	Attribute Based Access Control
ACAS	Assured Compliance Assessment Solution
ACC	Advanced Campaign Cell
ACL	Access Control List
ADDE	Aeronautical Digital Data Environment
ADE	Architecture Design Element
ADV	Application Development Services
AF9	Air Force Form 9
AIT	Active Information Technology
AMI	Amazon Machine Image
AMS	Aeronautical Migration System
AoA	Analysis of Alternatives
AOE	Aeronautical Obstruction Environment
API	Application Programming Interface
AR	Architecture Repository
ARH	Access Rights and Handling
ART	Agile Release Train
ASGB	Application Services Governance Board
ASP	Application Service Provider
ATLAS	Advanced, Technical, Leadership, Analytics Suite
ATO	Approval To Operate
ATP	Authorization To Proceed
AuthN	Authentication
AuthZ	Authorization
AWS	Amazon Web Services
AZ	Availability Zone
BIOS	Basic Input / Output System
BLOB	Binary Large Object
BOE	Body of Evidence
BYOL	Bring Your Own License
C&A	Certification and Accreditation
C2S	Commercial Cloud Services
CAC	Computer Access Card
CAMS	Culture, Automation, Measurement and Sharing
CAP	Cloud Access Point

Acronym	Description
CAST	Categorization and Selection TEM
CAT	Cloud Adoption Template
CCP	Common Control Provider
CD	Continuous Delivery
CDO	Chief Data Officer
CFM	Chargeback Financial Model
CI	Continuous Integration
CI/CD	Continuous Integration / Continuous Deployment
CIO	Chief Information Officer
CLIN	Contractor Line Item Number
CONOPS	Concept of Operations
COTS	Commercial-Off-The-Shelf
CND	Computer Network Defense
CNDSP	Computer Network Defense Service Provider
CRQ	Change Request
CSOC	Cyber Security Operations Center
CSP	Cloud Service Provider
CUI	Controlled Unclassified Information
DAFIF	Digital Aeronautical Flight Information File
DAO	Delegated Authorizing Official
DAR	Data at Rest
DAST	Dynamic Application Security Testing
DCOS	Data Center Operating System
DDR	Deactivation and Disposal Review
DevOps	Development Operations
DISA	Defense Intelligence Security Agency
DIT	Data in Transit
DNC	Digital Nautical Chart
DNI	Director of National Intelligence
DoD	Department of Defense
DoDAF	DoD Architecture Framework
DOS	Denial of Service
DR	Disaster Recovery
DSA	Data Services Architecture
DSP	Data Service Provider
E2C	Elastic Compute Cloud
EA	Enterprise Audit
EBS	Elastic Block Storage
EC2	Elastic Compute Cloud
ECDS	Enterprise Cross Domain Service
ECM	Enterprise Competency Model
EDH	Enterprise Data Header
ELA	Enterprise License Agreement
ELB	Elastic Load Balancer
EM	Entitlements Management

Acronym	Description
ERB	Engineering Review Board
ESE&I	Enterprise Systems Engineering & Integration
ESL	Enterprise Services List
ETL	Extract, Transform and Load
FedRAMP	Federal Risk and Authorization Management Program
FISMA	Federal Information Security Management Act
FM	Financial Management
FOC	Final Operational Capability
FOCI	Foreign Owned Controlled & Influenced
FOSS	Free or Open Source Software
FTE	Full Time Equivalent
FYDP	Future Years Defense Program
GEA	GEOINT Enterprise Architecture
GEOAxIS	GEOINT Access and Information Sharing
GEOINT	Geospatial Intelligence
GOTS	Government of the Shelf
GPOC	Government Point of Contact
GS	GEOINT Services
GSR	General Service Requirements
GUI	Graphical User Interface
GUIDE	Global Unique IDentifier for Everything
HIPAA	Health Insurance Portability and Accountability Act
IAA	Identity, Authentication and Authorization
IaaS	Infrastructure-as-a-Service
IAE	Integrated Analytic Environment
IAM	Identity and Access Management, or Identity Access Manager
IARC	Information Assurance Reference Catalog
IAST	Interactive Application Security Testing
IATO	Initial Authorization to Operate
IAW	In Accordance With
IC	Intelligence Community
ICD	Intelligence Community Directive
IC GovCloud	Intelligence Community Government Cloud
IC ISM	Intelligence Community Information Security Marking
IC ITE	Intelligence Community Information Technology Enterprise
ICMP	IC Marketplace
IC PAG	IC Program Architecture Guide
ICS	Intelligence Community Standard
IdAM	Identity Access Management
IEM	Infrastructure Event Management
IGCE	Independent Government Cost Estimate
IIL	Information Impact Level
IOC	Initial Operational Capability
IP	Internet Protocol
IPPBES	Intelligence Planning, Programming, Budgeting and Execution System

Acronym	Description
IS	Information Systems
ISM	Information Security Marking
ISO	International Standards Organization
ISP	Infrastructure Service Provider
IT	Information Technology
ITAR	International Traffic in Arms Regulations
ITIL	Information Technology Infrastructure Library
MIB	Mission Integration Board
J&A	Justification & Approval
JARM	Joint Architecture Reference Framework
JIE	Joint Information Environment
JSF	Joint StoreFront
JWICS	Joint Worldwide Intelligence Communications System
KMS	Key Management Service
LCD	Lowest Common Denominator
LDAP	Lightweight Directory Access Protocol
MASH	MachineShop
MUG	Mission Users Group
MVP	Minimum Viable Product/Process
NACL	Network Access Control List
NARA	National Archives and Records Administration
NAT	Network Address Translation
NCE	NGA Campus-East
NCW	NGA Campus West
NDAAS	NSG Data Analytic Architecture Service
NIST	National Institute of Science and Technology
NGA	National Geospatial-Intelligence Agency
NMF	NSG Metadata Foundation
NPE	Non-Person Entities
NSA	National Security Agency
NSG	National System for Geospatial Intelligence
NSS	National Security System
NTK	Need to Know
OCONUS	Outside the Contiguous United States
ODNI	Office of the Director of National Intelligence
ODS	OpenDataStore
OPR	Office of Primary Responsibility
OS	Operating System
PaaS	Platform-as-a-Service
PBR	Program Budget Review
PCF	Pivotal Cloud Foundry
PDP	Policy Decision Point
PE	Person Entities
PEP	Policy Enforcement Point
PHI	Personal Health Information

Acronym	Description
PII	Personal Identifying Information
PKI	Public Key Infrastructure
PfM	Portfolio Manager
PM	Program Manager
PMI	Project Management Institute
PMO	Program Management Office
POC	Point of Contact
POM	Program Objective Memorandum
PoR	Program of Record
PUM	Proper Use Memorandum
RBAC	Role-Based Access Control
RDP	Remote Desktop Protocol
RDS	Relational Database Services
REvAMP	Risk Evaluation Acceleration Management Process
RMF	Risk Management Framework
ROI	Return on Investment
S3	Simple Storage Service
SA	System Administrator
SaaS	Software-as-a-Service
SAFe	Scaled Agile Framework
SAM	Software Asset Management
SAP	Special Access Program
SAST	Static Application Security Testing
SC2S	Secret Commercial Cloud Service
SCA	Security Control Assessment
SCI	Sensitive Compartmented Information
SCTM	Security Controls Traceability Matrix
SFA	Foundation GEOINT Aeronautical Navigation Office
SGID	Set owner Group ID
SIDR	Security Impact Decision Request
SIN	Special Item Number
SME	Subject Matter Expert
SOP	Standard Operational Procedure
SOW	Statement of Work
SRB	Service Registry Board
SPID	Security Plan Identification Number
SSE	Server-side Encryption
SSH	Secure Shell
SSL	Secure Socket Layer
SSP	System Security Plan
STIG	Security Technical Implementation Guidance
STS	Secure Token Service
SUID	Set owner User ID
SWAP	Software Assurance Process
SWID	Software ID

Acronym	Description
T&Cs	Terms and Conditions
TDF	Trusted Data Format
TDO	Trusted Data Object
TDS	Topographic Data Store
TEM	Technical Exchange Meeting
TFDM	Topographic Features Data Management
TLS	Transport Layer Security
TMS	Target Management System
TST	Technical Services Taxonomy
UC2S	Unclassified Commercial Cloud Service
UDS	Unified Data Strategy
UFR	Unfunded Requirement
ULA	Unlimited License Agreement
URL	Uniform Resource Locators
UUID	Universally Unique Identifier
UX	User Experience
VANDL	Visualization Analytics Data Lake
VM	Virtual Machine
VPC	Virtual Private Cloud
VPN	Virtual Private Network
WebDVOF	Web Digital Vertical Obstruction File
WFS-T	Transactional Web Feature Service
XML	eXtensible Markup Language